

first order and Higher degree diff Eqn.

\* Solve  $y = p(x-b) + \frac{a}{p}$ ,  $p = \frac{dy}{dx}$

$$\frac{dy}{dx} = \cancel{p} = \cancel{p} + (x-b) \frac{dp}{dx} - \frac{a}{p^2} \cdot \frac{dp}{dx}$$

$$\Rightarrow \frac{dp}{dx} \left( x-b - \frac{a}{p^2} \right) = 0$$

$$\frac{dp}{dx} = 0$$

$$\Rightarrow \boxed{p = c}$$

General sol :-  $y = c(x-b) + \frac{a}{c}$

$$x-b - \frac{a}{p^2} = 0$$

$$p^2(x-b) = a$$

$$p = \pm \sqrt{\frac{a}{x-b}}$$

→ Singular solution

• Clairaut's equation

$$y = px + f(p)$$

$$\frac{dy}{dx} = \cancel{p} = \cancel{p} + x \frac{dp}{dx} + f'(p) \frac{dp}{dx}$$

$$\Rightarrow \frac{dp}{dx} (x + f'(p)) = 0$$

$$\frac{dp}{dx} \Rightarrow \underline{p = c}$$

$$\rightarrow x + f'(p) = 0$$

Singular solution ←

Solution is  $y = cx + f(c)$ .

question :- Solve  $y = px + \sqrt{a^2 p^2 + b^2}$

Solution :- This is Clairaut's Eqn.

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we put  $P=C$   
 Solution is  $y = cx + \sqrt{a^2c^2 + b^2}$

• Equations, which are Reducible to Clairaut's Eqn.

$$y^2 = pxy + f\left(\frac{py}{x}\right)$$

we put  $v = y^2$   $u = x^2$

$$\frac{dv}{dy} = 2y$$

$$\frac{du}{dx} = 2x$$

$$P = \frac{dv}{du} = \frac{dv}{dy} \cdot \frac{dy}{dx} \cdot \frac{dx}{du} = 2y \cdot P \cdot \frac{1}{2x} = \frac{py}{x}$$

Clairaut's Eqn  
 $y = px + f(p)$

$$v = x^2 \left(\frac{py}{x}\right) + f(P) \Rightarrow \boxed{v = uP + f(P)}$$

↓  
This is Clairaut Eqn.

Solution is (by putting  $P=C$ )  $\Rightarrow v = uC + f(C)$

Solution of given Eqn is  $\Rightarrow y^2 = x^2C + f(C)$

Question :- Solve  $x^2y^2 = px^3y + p^2y^2 + pny + 9x^2$

$$y^2 = pxy + \frac{p^2y^2}{x^2} + \frac{py}{x} + 9$$

we substitute

$$y^2 = v$$

$$2y = \frac{dv}{dy}$$

$$x^2 = u$$

$$2x = \frac{du}{dx}$$

h

$$ay = \frac{u}{dy} dx$$

$$P = \frac{dv}{du} = \frac{dv}{dy} \cdot \frac{dy}{dx} \cdot \frac{dx}{du} = \frac{py}{x}$$

The eqn reduced to  $v = uP + P^2 + P + 9$

This is Clairaut's eqn,

Solution  $v = uc + c^2 + c + 9$

Solution of given eqn  $y^2 = x^2c + c^2 + c + 9$

• Solving linear diff Eqns with constant coefficients.

$$y'' + ay' + by = 0, \quad a, b \text{ are constants.}$$

Example :- Solve  $y'' - 6y' + 8y = 0$  — ①,  $y(0) = -2, y'(0) = 6$

Solution :- Let  $y = e^{rx}$  be solution of ①  
 $y' = re^{rx}, \quad y'' = r^2 e^{rx}$

we get

$$r^2 e^{rx} - 6re^{rx} + 8e^{rx} = 0$$

$$e^{rx} (r^2 - 6r + 8) = 0$$

ch. eqn

$$\leftarrow r^2 - 6r + 8 = 0 \Rightarrow r = 2, 4$$

$e^{2x}, e^{4x}$

Complete solution is  $y = C_1 e^{2x} + C_2 e^{4x}$

\* Let  $y^n + a_1 y^{n-1} + a_2 y^{n-2} + \dots + a_{n-1} y = 0$  be linear diff Eqn with constant coefficients of order  $n$ .

Ch Eqn

$$x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} = 0$$

polynomial of degree  $n$

Case I :- Let Ch Eqn has  $n$  distinct real roots say  $r_1, r_2, \dots, r_n$

Solution of Diff Eqn is

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x} + \dots + C_n e^{r_n x}$$

• Solve (1)  $y'' - k^2 y = 0$ ,  $k \in \mathbb{R}$  Ch Eqn  $x^2 - k^2 = 0$   
 $r = \pm k, k \in \mathbb{R}$

Solution is  $y = C_1 e^{kx} + C_2 e^{-kx}$

(2)  $y'' - 8y' + 16y = 0$  — (★)

Ch Eqn

$$x^2 - 8x + 16 = 0 \Rightarrow x = 4, 4$$

one solution is  $e^{4x} = y_1$

$$y_2 = u(x) e^{4x}$$

$$y_2' = 4u(x)e^{4x} + u'(x)e^{4x}$$

$$y_2'' = 16u(x)e^{4x} + 8u'(x)e^{4x} + u''(x)e^{4x}$$

$\therefore y_2$  is solution of (★)  $\dots \dots \dots 2u'(x)e^{4x} - 8u(x)e^{4x}$

$\therefore y_2$  is solution of  $\textcircled{A}$

$$\cancel{16u(x)e^{4x} + 8u'(x)e^{4x} + u''(x)e^{4x} - 32u(x)e^{4x} - 8u'(x)e^{4x}} + \cancel{16u(x)e^{4x}} = 0$$

$$u''(x)e^{4x} = 0 \Rightarrow u''(x) = 0$$

Put  $\phi(x) = u'(x)$

$$\Rightarrow \phi'(x) = 0 \Rightarrow \phi(x) = C_1$$

$$u'(x) = C_1 \Rightarrow u(x) = C_1 x + C_2$$

We take  $u(x) = x$

$$y_2(x) = xe^{4x}$$

Complete solution:  $y = C_1 e^{4x} + C_2 x e^{4x}$

\* Consider the linear diff Eqn with constant coefficients.

$$y^n + a_{n-1}y^{n-1} + a_{n-2}y^{n-2} + \dots + a_1 y = 0$$

Ch Eqn

$$x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1 = 0$$

let roots of Ch Eqn are

$$\underbrace{r_1, r_1, \dots, r_1}_R \text{ times}, r_2, r_3, \dots, r_{n-R+1}$$

Solution is

$$y = C_1 e^{r_1 x} + C_2 x e^{r_1 x} + \dots + C_R x^{R-1} e^{r_1 x} + C_{R+1} e^{r_2 x} + C_{R+2} e^{r_3 x} + \dots + C_n e^{r_{n-R+1} x}$$

question:-

$$y'' + k^2 y = 0, \quad k > 0$$

.2  $-k^2$

question:-

$$y'' + k^2 y = 0, \quad k \neq 0$$

Ch. Eqn

$$r^2 + k^2 = 0$$

$$\Rightarrow r^2 = -k^2$$

$$\Rightarrow r = \pm ki$$

Solution:-

$$y = C_1 e^{kix} + C_2 e^{-kix}$$

Euler's formula:-  $e^{i\theta} = \cos\theta + i\sin\theta$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

$$y = C_1 (\cos(kx) + i\sin(kx)) + C_2 (\cos(kx) - i\sin(kx))$$

$$y = (C_1 + C_2) \cos kx + i(C_1 - C_2) \sin kx$$

$$y = C'_1 \cos kx + C'_2 \sin kx$$

\* If characteristic Eqn has complex roots  
say  $a+ib, a-ib$   
Then solution

$$y = e^{ax} (C_1 \cos bx + C_2 \sin bx)$$

\* Solve

$$y'' - 6y' + 13y = 0$$

Ch. Eqn

$$r^2 - 6r + 13 = 0$$

$$r = 3 \pm 2i$$

$$y = C_1 e^{(3+2i)x} + C_2 e^{(3-2i)x}$$

$$y = e^{3x} (C_1 e^{2ix} + C_2 e^{-2ix})$$

$$y = e^{3x} (C_1 e^x + C_2 e^{-x})$$

$$y = e^{3x} (C_1' \cos 2x + C_2' \sin 2x)$$

o Solve  $y^{IV} + 2y'' + y = 0$

Solution

Ch Eqn

$$r^4 + 2r^2 + 1 = 0$$

$$(r^2 + 1)^2 = 0$$

$$r^2 + 1 = 0$$

$$r^2 + 1 = 0$$

$$r = \pm i$$

$$r = \pm i$$

Solution is

$$y = C_1 \cos x + C_2 \sin x + C_3 x \cos x + C_4 x \sin x$$